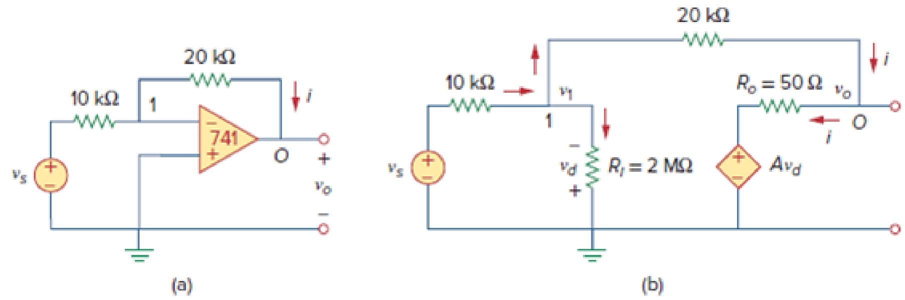
Problem

Repeat Example 5.1 using the ideal op amp model.

Example 5.1:

A 741 op amp has an open-loop voltage gain of 2 × 105, input resistance of 2 MΩ, and output resistance of 50 Ω. The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s. Determine current i when $v_s = 2$ V.

Figure. 5.6: (a) original circuit, (b) the equivalent circuit.



Solution:

Using the op amp model in Fig. 5.4, we obtain the equivalent circuit of Fig. 5.6(a) as shown in Fig. 5.6(b). We now solve the circuit in Fig. 5.6(b) by using nodal analysis. At node 1, KCL gives

$$\frac{v_s - v_1}{10 \times 10^3} = \frac{v_1}{2000 \times 10^3} + \frac{v_1 - v_o}{20 \times 10^3}$$

Multiplying through by 2000 × 103, we obtain

$200v_s = 301v_1 - 100v_o$

Or

$$2v_s \simeq 3v_1 - v_o \quad \Rightarrow \quad v_1 = \frac{2v_s + v_o}{3} \qquad (5.1.1)$$

At node O,

$$\frac{v_1 - v_o}{20 \times 10^3} = \frac{v_o - Av_d}{50}$$

But $v_d = -v_1$ and $A = 200{,}000$. Then

$$v_1 - v_o = 400(v_o + 200{,}000v_1) \qquad (5.1.2)$$

Substituting v_1 from Eq. (5.1.1) into Eq. (5.1.2) gives

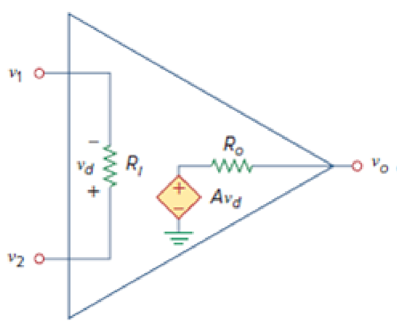
$$0 \simeq 26{,}667{,}067v_o + 53{,}333{,}333v_s \quad \Rightarrow \quad \frac{v_o}{v_s} = -1.9999699$$

This is closed-loop gain, because the 20-kΩ feedback resistor closes the loop between the output and input terminals. When $v_s = 2$ V, $v_o = -3.9999398$ V. From Eq. (5.1.1), we obtain $v_1 =$ 20.066667 μV. Thus,

$$i = \frac{v_1 - v_o}{20 \times 10^3} = 0.19999 \text{ mA}$$

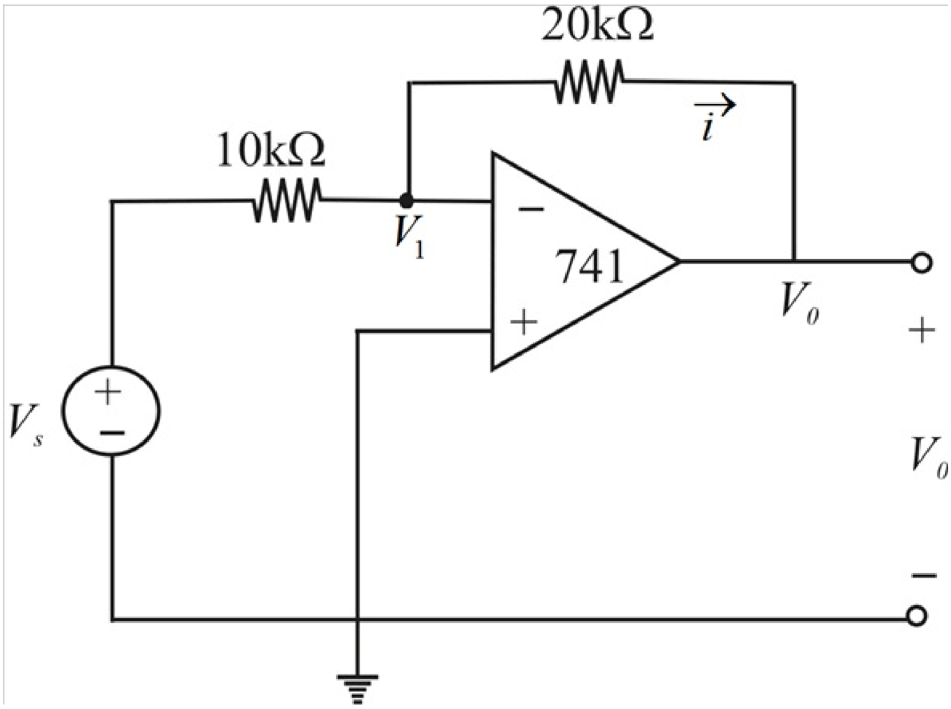
It is evident that working with a nonideal op amp is tedious, as we are dealing with very large numbers.

Figure. 5.4: The equivalent circuit of the nonideal op amp.



Step 1 of 4

Considering the circuit



Step 2 of 4

By applying KCL node (1) we have

$$\frac{V_1 - V_s}{10\text{k}} + \frac{V_1 - V_0}{20\text{k}} = 0$$

As the OP amp is ideal such that

$$V_1 = 0$$

Thus the equation we have by substituting $V_1 = 0$

$$\frac{0 - V_s}{10\text{k}} + \frac{0 - V_0}{20\text{k}} = 0$$

$$-\frac{V_s}{10\text{k}} - \frac{V_0}{20\text{k}} = 0$$

$$-\frac{V_0}{20\text{k}} = \frac{V_s}{10\text{k}}$$

$$-0.05V_0 = 0.1V_s$$

$$\frac{V_0}{V_s} = \frac{-0.1}{0.05}$$

$$\boxed{\frac{V_0}{V_s} = -2}$$

Step 3 of 4

In order to obtain i we have

$$i = \frac{V_1 - V_0}{20\text{k}}$$

Given $V_s = 2\text{V}$

$$\frac{V_0}{V_s} = -2$$

$$\frac{V_0}{2} = -2$$

$$V_0 = -4\text{V}$$

Step 4 of 4

Substituting this V_0 in the equation '1'

$$i = \frac{V_1 - V_0}{20\text{k}}$$

$$i = \frac{0 + 4}{20\text{k}}$$

$$i = \frac{1}{5\text{k}}$$

$$\boxed{i = 0.2 \text{ mA}}$$